

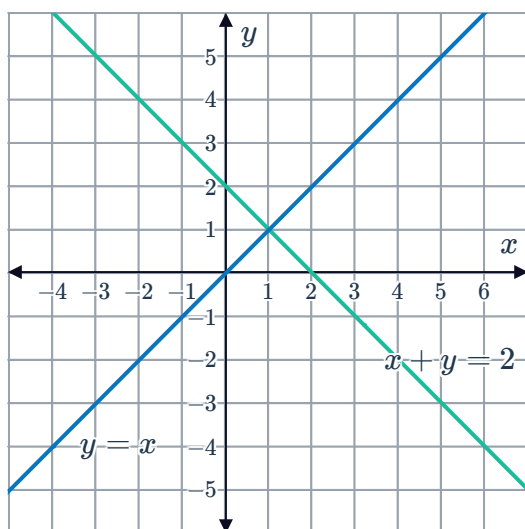
Areas under multiple curves



1

a

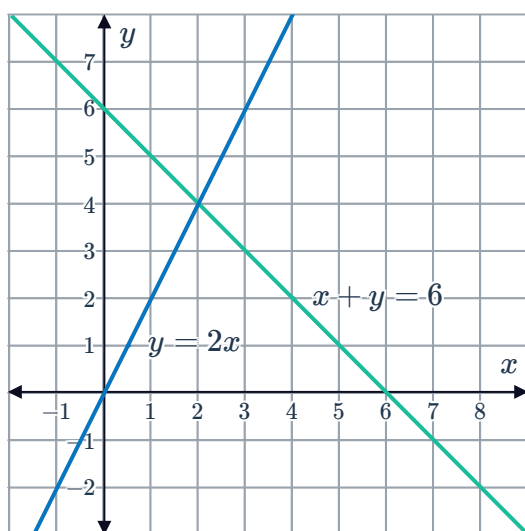
i



ii 1 unit^2

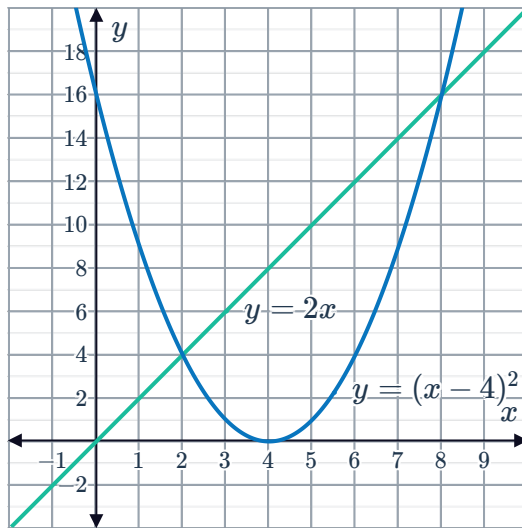
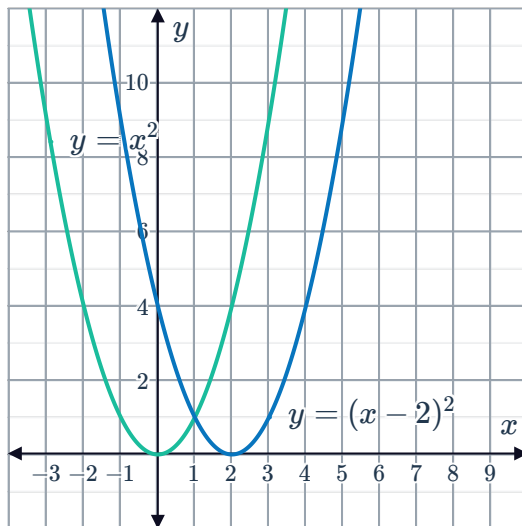
b

i



ii 12 units^2

2

a
iii $x = 2, 8$ iii $\frac{20}{3} \text{ units}^2$ b
iii $x = 1$ iii $\frac{2}{3} \text{ units}^2$

3

a $\frac{14}{3} \text{ units}^2$ b $\frac{103}{12} \text{ units}^2$ c $\frac{44}{3} \text{ units}^2$ d $\frac{11}{24} \text{ units}^2$ 4 $\frac{7}{6} \text{ units}^2$ 5 $\frac{25}{3} \text{ units}^2$ 6 $\frac{3}{2} - \frac{1}{2}e^{-4} - e^{-2} \text{ units}^2$

7 $\frac{89}{3} \text{ units}^2$



8 $\frac{50}{3} \text{ units}^2$

9 a 8 units^2 b 16 units^2 c $2 : 1$

10 $k = 2$

11 $4 - \pi \text{ units}^2$

12 a $x = 9$ b $\left(3e^{\frac{3}{2}} + 6\right) \text{ units}^2$

13 a $(e^3 - 1) \text{ units}^2$ b $y = e^3 x - 2e^3$ c $\left(\frac{1}{2}e^3 - 1\right) \text{ units}^2$

Areas between curves

14 30 units^2

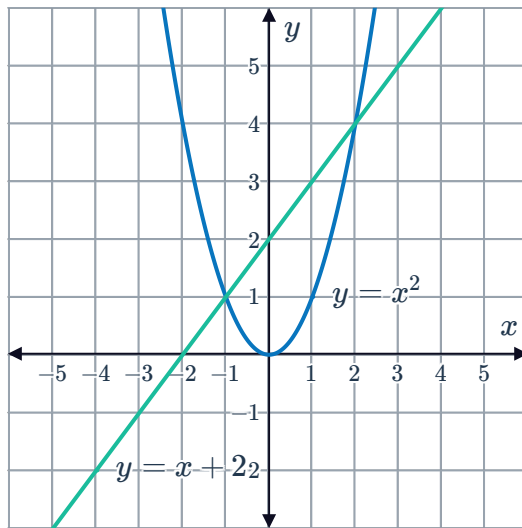
15 $\frac{28}{3} \text{ units}^2$

16 $\frac{4}{15} \text{ units}^2$

17 a $x = 0, 4, 9$ b $\frac{2521}{12} \text{ units}^2$ c $\frac{100}{3} \text{ units}^2$



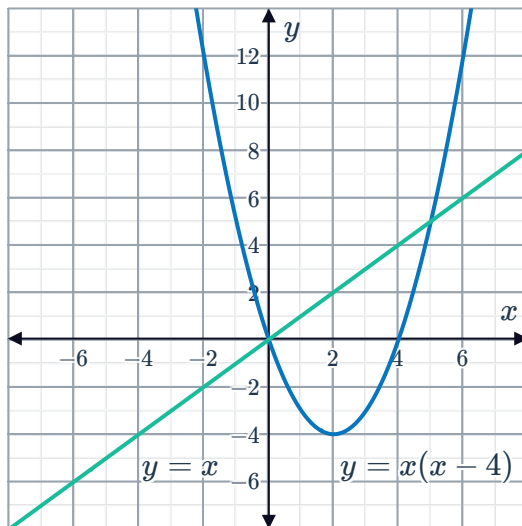
a
i



ii $x = -1, 2$

iii $\frac{9}{2}$ units²

b
i

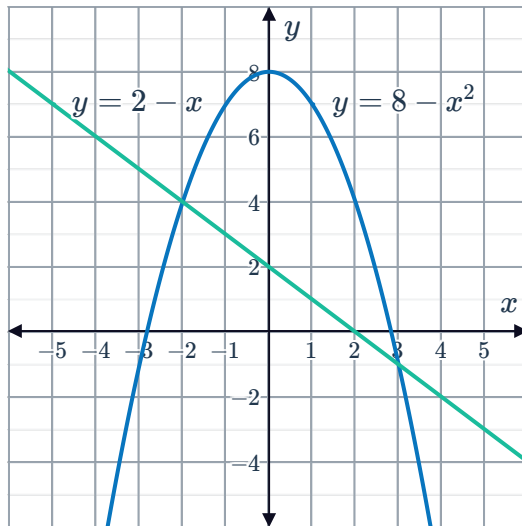


ii $x = 0, 4$

iii $\frac{125}{6}$ units²

c

i



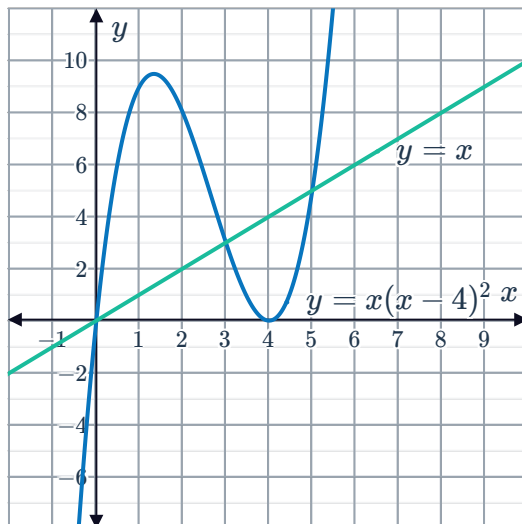
ii

$$x = -2, 3$$

iii $\frac{125}{6} \text{ units}^2$

d

i



ii

$$x = 0, 3, 5$$

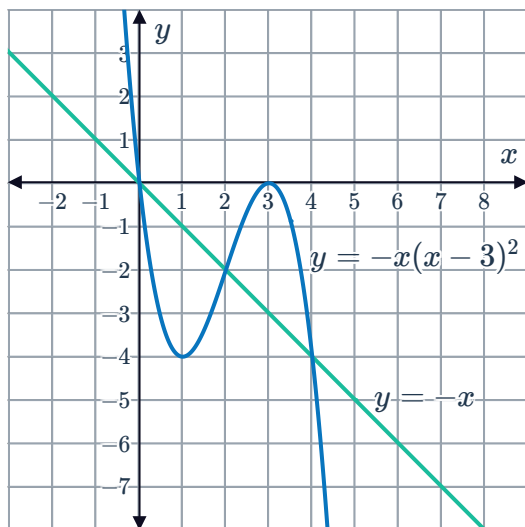
iii $\frac{253}{12} \text{ units}^2$

e

i



ii $x = 0, 2, 4$



iii 8 units^2

19

a

i $x = -4, 6$

ii $\frac{1000}{3} \text{ units}^2$

b

i $x = 4, 5, 6$

ii $\frac{1}{2} \text{ units}^2$

c

i $x = -1, 3, 4$

ii $\frac{131}{4} \text{ units}^2$

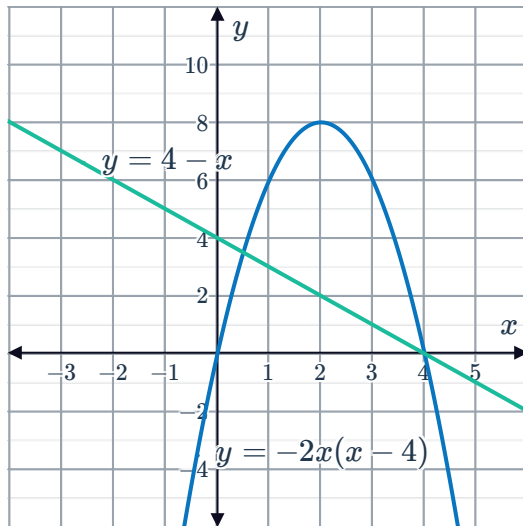
d

i $x = 5, -3$

ii $\frac{256}{3} \text{ units}^2$

20

a



b

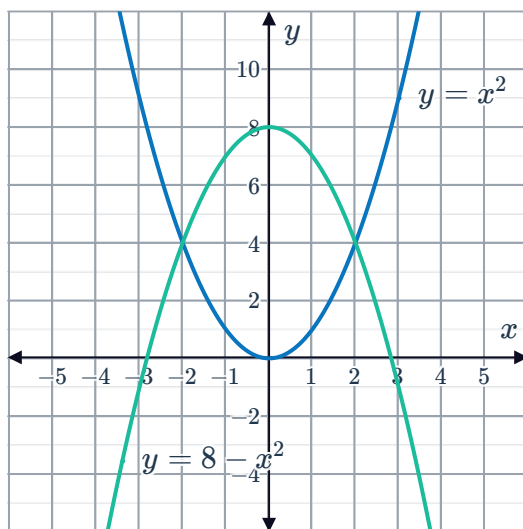
$$x = \frac{1}{2}, 4$$

c $\frac{343}{24} \text{ units}^2$

d $\frac{23}{24} \text{ units}^2$

21

a



b $x = -2, 2$

c $\frac{64}{3} \text{ units}^2$

d 8.8 units^2

22

a

i Yes

ii Yes

iii No

iv Yes

v No

vi No

b 4 units^2

c $2 : 1$

23 $\frac{2}{\sqrt{2}} - 1 \text{ units}^2$

$$\frac{1}{4} \text{ units}^2$$



$$25 \frac{1}{4} (\sqrt{2} - 1) \text{ units}^2$$

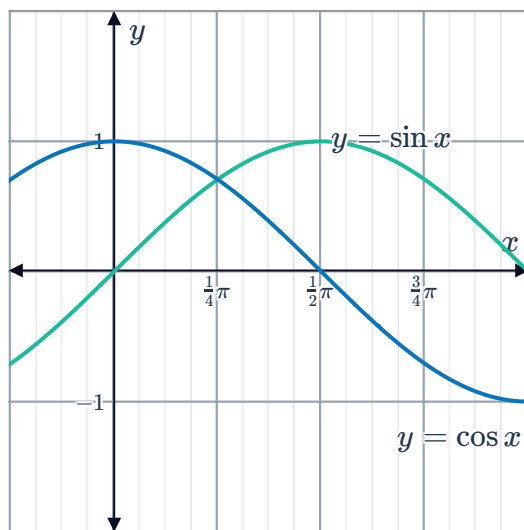
$$26 \ 11.89 \text{ units}^2$$

$$27 \ 4\sqrt{2} + 4 \text{ units}^2$$

$$28 \ \frac{10\pi^2}{9} + 3 \text{ units}^2$$

29

a

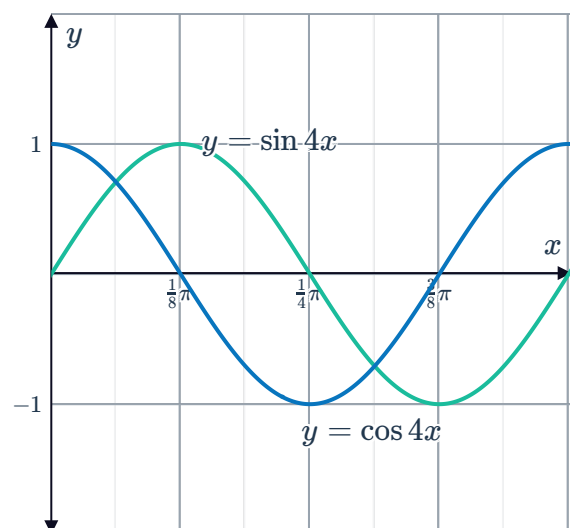


b $2\sqrt{2} \text{ units}^2$

30

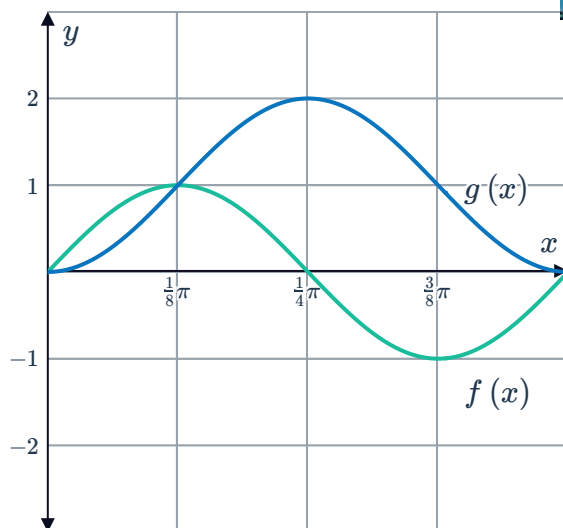
a $x = \frac{\pi}{16}, \frac{5\pi}{16}$

b



c $\frac{\sqrt{2}}{2} \text{ units}^2$

31 a



b $0, \frac{\pi}{8}, \frac{\pi}{2}$

c $\frac{\pi}{2}$

d $\frac{\pi}{4} + 1 \text{ units}^2$

32

a 4 units^2

b $e^3 + e^{-3} - 2 \text{ units}^2$

c $\frac{2}{5}e^{10} + \frac{2}{5}e^{-10} - \frac{4}{5} \text{ units}^2$

d $3 + e^{-2} \text{ units}^2$

33

a $x = 3$

b $2e^3 - 17 \text{ units}^2$

34

a $x = 0, 1$

b $\frac{1}{3} \text{ units}^2$

c

n	2	3	5	8
Area	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{7}{9}$

d The area between the curves $f(x) = x^n$ and $g(x) = \sqrt[n]{x}$ for positive integer values of n is $\frac{n-1}{n+1}$.

e No solution provided.

35

a -22

b 34

c 13 units^2

d 5 units^2

e 29 units^2

36

a 16

b -10



c 1 unit²

d 20

37

a 18 units²

b $\frac{22}{3}$ units²

c $a = \frac{182}{75}$